

Are Lexical Representations Separable? Item and Category in Learned Distributed Representations

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Abstract

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1 Introduction

2 Experiment 1: Artificial Languages

The goal of Experiment 1 is to validate our measurement on an artificial language whose structure we control, and to examine how a learner represents that structure.

2.1 Dataset

In order to examine the separability of linguistic representations, we constructed an artificial language which comprises a probability distribution over a vocabulary of 600 tokens. Every observation is a combination of a **form** (one of 60), a **class** (one of 4), and a **context** (drawn from a pool of 400). The motivation for this setup is to give each word a naturalistic environment: the word occurs in a context that favors a specific class-membership (e.g., *walk* as a verb occurs in a different context than *walk* as a noun). Each form is observed 24 times in each class, every time in a different context, mirroring the patterns of human language. Consider *walk* used as a verb, in one of those contexts. Four things determine the distribution over the next token. *Walk* itself contributes the same amount regardless of the class it is used in, and the verb class also contributes the same amount for every item. The third contribution is what is specific to *walk*-as-a-verb over and above those two: every form and every class has its own random vector, and this term multiplies *walk*'s by the verb's, entry by entry. The context contributes the rest, which is noise as far as our question goes, though the model still has to encode it in order to correctly predict the distribution. The four are weighted, summed and passed through a softmax:

$$P(\text{next token} \mid f, c, x) = \text{softmax} \left(s \left[w_{\alpha} \text{item}(f) + w_{\beta} \text{class}(c) + w_{\gamma} \text{int}(f, c) + w_{\text{ctx}} \text{context}(x) \right] \right) \quad (1)$$

Each term is scaled to unit length before it is weighted, so changing the weights changes the mix of the three effects without changing how much signal there is in total, and the task stays equally hard everywhere in the factorial.

The weights w_{α} , w_{β} and w_{γ} in Equation 1 control how much of the true distribution each of the three effects contributes, and we cross them factorially over $\{0, 0.5, 1, 2\}$: zero removes an effect from the language altogether, and the other three values make an effect half as strong as another, equal to it, or twice it. Similarly, w_{ctx} governs how much the context contributes, so we cross those languages with three context weights: $\{0.5, 2.0, 6.0\}$, ranging from context being a minor part of the true distribution to the dominant one. We train a model on each of the resulting languages at hidden widths of 8, 16, 32, 128, with either a linear or a ReLU activation. The whole grid is then run with five random seeds, each redrawing the language and reinitializing the model, giving 7,560 runs in all, of which 99.7% trained to a loss plateau.

Entanglement is guaranteed under a specific subset of the languages. The structure needs 16 dimensions, so its ratio to the hidden width, which we call **crowding**, runs from 0.125 to 2.0. Below one there is room to give every component its own axis, so any entanglement was chosen; above one there are more components than dimensions to put them in, so they cannot all have their own axis and entanglement is forced (Elhage et al., 2022).

2.2 Method

All code and analyses in this paper can be found here: <https://github.com/znhoughton/>

representation-separability.

2.2.1 The learner

The learner model is a multi-layer perceptron (MLP) with a single hidden layer of width d and either a linear or a ReLU activation function. The model takes as its input three pieces of information: form, class, the context. Each has its own embedding, and the three are concatenated and passed through the model which then outputs a probability distribution over the vocabulary (Figure 1). The training objective is thus to minimize cross-entropy between the model’s predicted distribution over the vocabulary and the true distribution.

2.2.2 Decomposing a representation

We measure separability by decomposing a word-in-context representation into a lexical component, a class component, and an idiosyncratic component. The representation is the MLP’s hidden layer, which for a given form and class varies across the contexts it is observed in, so let $M[i, c] \in \mathbb{R}^d$ be the mean of those hidden states for item i used at class c . We decompose those item-by-class means by centering, so that:

$$M[i, c] = \mu + \alpha(i) + \beta(c) + \gamma(i, c), \quad (2)$$

where μ is the grand mean; $\alpha(i)$, item i ’s mean across classes minus μ , carries the lexical information; $\beta(c)$, the mean across items at class c minus μ , carries the class information; and $\gamma(i, c)$ carries the idiosyncratic information, what item i at class c holds that neither marginal accounts for (Figure 2). Because every item is observed in every class with equal weight, neither marginal absorbs part of the other. The four terms add back to $M[i, c]$ exactly, and no two of them vary together across the item-by-class means, so the squared lengths of the three that vary sum to the total and we report each as a share of it.

Each of these averages is taken over a sample of observations that differ only in context, so each carries some of the context term of Equation 1 as noise. We remove it by dividing the observations behind each average into two even-sized splits, running the decomposition on each split separately, and estimating each component’s size as the dot product of its two split estimates, which is large when the splits agree about the component and near zero when they do not. It is the contexts that are divided and not the words, so every word appears

in both splits, and the dot product compares two estimates of the same word computed from different sentences. As a result, noise that differs between the splits is independent and cancels.

For each representation we report two measurements. The first measure is the fraction of the representation accounted for by lexical, class and idiosyncratic information: each component’s squared length, summed over its vectors and divided by the total, so that the three fractions sum to one. The second measures whether those components occupy the same directions. In order to do so, we project the item vectors onto the class direction and record the fraction of their squared length that survives, and we project the interaction onto the smallest subspace containing both the item and the class vectors (i.e., the span of the two subspaces). A representation is separable if a word’s representation is the sum of an item part and a class part, each on its own axis. Separability can fail in two ways: the parts may be non-orthogonal, so that projecting out the class also removes item information, or there may be a third component that is neither, whatever its orientation. The question is accordingly not whether class information is present, but how it is carried: on a single direction shared across words, or on a separate direction for each of them.

To quantify whether a size is meaningful, we compare each against zero. Size is estimated by taking the dot product of two independent split estimates of the same component, so noise that differs between the splits cancels out and its expected value is zero when the component is absent, which is what makes zero the null. We repeat the measurement over two hundred re-samplings to estimate how much the size varies with which observations fall in which split, and we call a size meaningful when its confidence interval excludes zero. On representations with a component planted at exactly zero, this flags an effect about 5% of the time (Appendix A), the false-positive rate expected of a 95% confidence interval.

In order to quantify whether the overlap is meaningful, we measure the overlap an effect has with a random subspace. We do so by holding the effect exactly as measured and projecting it onto a random subspace of the same rank, two hundred times, which leaves the effect untouched and varies only the subspace. These two hundred projections give us a distribution of overlaps that an unrelated subspace would yield. An overlap lower than this baseline indicates meaningful orthogonality; an



Figure 1: What the learner sees. Every form is observed 24 times at each class, each time in a different context; that variation across contexts is what the measurement later splits in two.

overlap larger than it indicates meaningful non-orthogonality. We present each overlap as a multiple of the median of this distribution, together with how many measurements fall outside its 95% interval. This gives both the size of the effect and how consistently it holds across measurements.

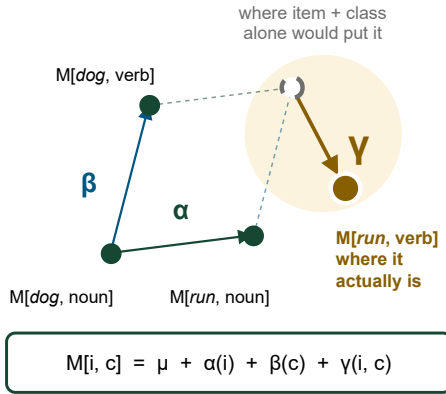


Figure 2: The four item-by-class means formed by two items in two classes. Under an item part plus a class part, three corners fix the fourth and the parallelogram closes. The arrow is $\gamma(i, c)$, the displacement between the fourth mean and the point that additive account places it at, with length $\|\gamma(i, c)\|$.

It’s important to note that the weights in Equation 1 describe the distribution the model is trained to match, while the measure reads the representation it forms, so the two sets of numbers are not on a common scale: a learner handed a language that is 85% interaction can represent it with the interaction taking a far smaller share. Appendix A reports a separate check in which components were planted directly into a representation and recovered almost exactly, which is what licenses reading the results below as facts about the learner rather than about the measure. A run contributes to the results only if at least one of the three components has a re-split interval excluding zero; where none does, the three shares divide noise and whichever lands

slightly positive takes all of it.

2.3 Results

Results are shown in Figure 3; the effects of crowding and of the language’s other properties are in Appendix B.

The linear condition reads zero only where the language is additive. Across the 2,880 runs whose language contained an interaction, the linear model’s interaction interval excludes zero in 36% of them, against 77% for ReLU; where no interaction was planted, the rates are 2% and 13%. The linear model has no parameter depending on the form and the class together, yet represents an interaction anyway: a word-in-a-class is averaged over its own contexts, and that average is neither a fact about the word nor a fact about the class. The models with a ReLU activation function surprisingly represent an interaction even in cases where the language is additive. One potential explanation is that the model only ever encounters words in specific contexts; the additive structure of the language is visible across the corpus rather than in any single observation, and the model does not seem to abstract away the interaction completely.

As the hidden width grows, the added room seems to be allocated toward representing the interaction. From a hidden width of 16 to a width of 128, the interaction’s share of the item-by-class structure rises from 0.002 to 0.068, while the item effect falls from 0.58 to 0.47 and the class effect holds steady at around 0.43. Regressing each measured share on the share planted in the language, still under a ReLU and at width 128, gives a slope of 0.63 for the item effect and 0.46 for the class effect, against 0.15 for the interaction. In the linear model the same width is spent elsewhere: the item effect rises from 0.61 to 0.72 while the interaction stays at 0.006. This suggests that the interaction is gated by crowding: it is represented at a lower rate

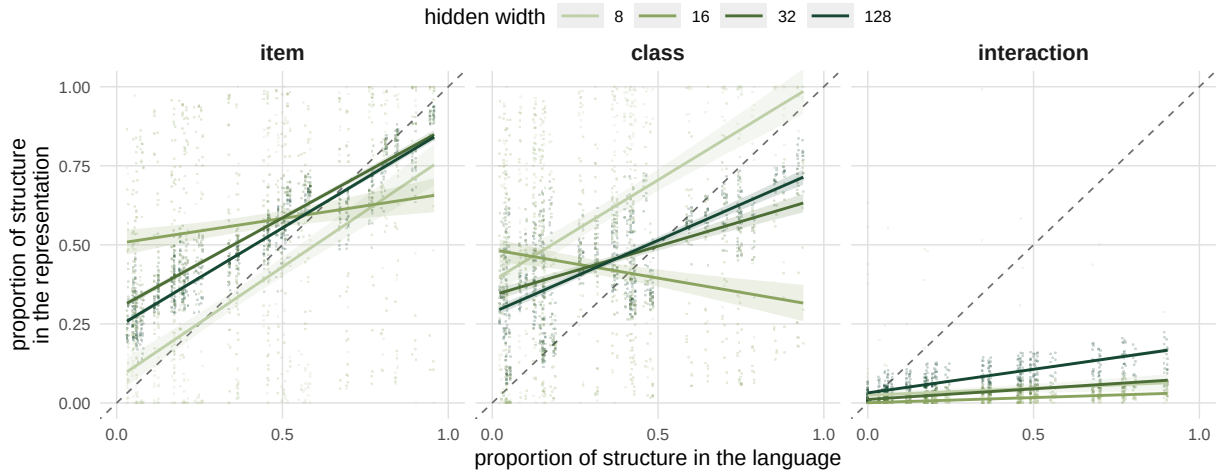


Figure 3: What the learner represents of each component, against how much of that component the language held, under ReLU. Each faint point is one run. Horizontally it sits at that component’s proportion of the structure in the language, with context set aside; vertically at its proportion of the structure in the representation. Both are proportions of the same three components, so the two are on a common scale. Lines are least-squares fits through the runs at each hidden width, shaded with a 95% confidence band on the fit; how far the runs themselves scatter is shown by the points, not by that band. The dashed diagonal is where a representation would sit that reproduced the language’s composition exactly: above it a component is over-represented relative to the language, below it under-represented. Runs in which none of the three components had a re-split 95% interval excluding zero are not plotted, since the three shares then divide noise; all such runs are at width 8, and their number is given in the text.

than either main effect, and it is the only one of the three still growing at the widest representation we train.

Additionally, at lower dimensions the representation seems to be more variable. This is not particularly surprising since each model run has to decide which information to represent with only a small number of dimensions. As a result, from run to run the model makes different decisions with respect to which information to represent in its limited dimensionality yielding vastly different representations of the same language. For example, at a width of 8 two runs of the same language differ by 0.47 in the item effect and 0.52 in the class effect, roughly half the available range; by a width of 32 that has fallen to 0.08 and 0.07, and it falls further at 128.

Finally, crowding also governs the amount of overlap between the item and class effects (Appendix B). With crowding well below one, ReLU’s item-class overlap is 0.0092 against a chance level of 0.0297; at a crowding of 1 it is 0.402, roughly 1.4 times chance, and at a crowding of 2 it is 0.512. The linear condition stays at or below chance throughout.

3 Experiment 2: Language Models

Experiment 1 established that if the language a learner model encounters is not purely additive,

the model will represent the language in a non-separable manner, the extent of which is governed by the size of the model. Experiment 2 applies the same decomposition to the contextual representations of pretrained language models, across three linguistic categories: part of speech, grammatical role, and metaphor.

3.1 Dataset

In order to examine the separability of linguistic representations in models, we examined three linguistic categories: part of speech, grammatical role, and metaphor. For part of speech and grammatical role we concatenate six general-domain English Universal Dependencies treebanks (Nivre et al., 2020): EWT (Silveira et al., 2014), GUM (Zeldes, 2017), LinES (Ahrenberg, 2007), ParTUT (Sanguinetti and Bosco, 2015), GENTLE (Aoyama et al., 2023) and PUD (Zeman et al., 2017), giving 41,095 sentences and 706,323 tokens. A form must be attested in both classes often enough to estimate a mean at each, and a single treebank yields too few of such forms. For metaphor we use the VUA20 corpus (Leong et al., 2020; Steen et al., 2010), restricted to nouns, verbs, adjectives, and adverbs in VUA’s own annotation, which supplies 88,509 labeled tokens. We measure two model families at three matched scales: OPT-BabyLM,

trained on developmentally plausible quantities of child-available language (Houghton et al.; Warstadt et al., 2023), and Pythia, trained on the Pile (Biderman et al., 2023), at 125M/160M, 350M/410M and 1.3B/1.4B parameters, with widths of 768, 1024 and 2048. Appendix C gives parameter counts and training-token estimates.

3.2 Stimuli

Our stimuli comprise three linguistic categories: part of speech (NOUN versus VERB), grammatical role (*nsubj* versus *obj*, restricted to nouns), and metaphor (literal versus figurative), each taken from its corpus’s own annotation. For each category we select items that occur in both of its classes, such as *walk*, which occurs as both a noun and a verb. An item is a single inflected word form, not a lemma, so different inflections of one lemma are separate items: *walk* and *walking* are both items, each included only if it is itself attested in both classes. Pooling them under one lemma would collect *walk*, *walks*, *walked* and *walking* into a single item, and the two classes would then be read at different tokens, which the embedding layer could then separate before any contextualization has occurred. Capitalization is ignored when occurrences are grouped, but not in the text the models read, so a sentence-initial *Walk* counts toward the item *walk* and is tokenized as it appears there.¹

An item is included only if it occurs at least ten times in each class; forms occurring in only one are excluded, which makes part of speech the most restrictive of the three. This yields 78 items for part of speech, 49 for grammatical role and 180 for metaphor, with median counts per class running from 15 to 26 (Table 1), inside the ten-to-one-hundred range the validation covers (Appendix A).

3.3 Method

The decomposition, the two reported quantities and the replication logic are identical to Experiment 1. For every token we take the hidden state of the word’s **last subword** at every layer, from the embeddings through the final block, aligning subwords to corpus tokens with the tokenizer’s word

¹OPT-BabyLM’s tokenizer is uncased, so this choice has no affect for that family of models. On the other hand, Pythia’s tokenizer is not uncased, however lowercasing the input would be the larger distortion, since a sentence-initial word is capitalized in written English which the model was trained on.

class	item types	median tokens
part of speech (noun / verb)	78	18 / 18
grammatical role (subject / object)	49	15 / 19
metaphor (literal / figurative)	180	26 / 21

Table 1: Stimuli for the three linguistic categories, with the two classes of each in parentheses. *Item types* is how many word forms occur in both of them at least ten times; forms occurring in only one are excluded. *Median tokens* is how often a typical form occurs in each of its two classes, in the order they are listed.

alignment.²

Transformer hidden states contain a few *rogue* dimensions whose variance exceeds the rest by two to three orders of magnitude (Timkey and van Schijndel, 2021), and these would otherwise dominate any subspace estimated from the raw scatter, so we divide each dimension by its standard deviation before decomposing. The measure is accordingly invariant to per-dimension rescaling but not to arbitrary linear reparameterization, which we return to in Appendix D.

3.4 Results

The results for the final layer are reported in Table 13; the layer-by-layer results are shown in Figure 4.

The item effect is the largest component in all three categories, running from 0.656 to 0.829 for part of speech, 0.828 to 0.902 for grammatical role, and 0.916 to 0.976 for metaphor. What differs between them is how the remainder divides. For grammatical role the class effect, at 0.047 to 0.078, is comparable to the interaction at 0.041 to 0.094; for part of speech the interaction, at 0.091 to 0.245, runs well above the class effect at 0.057 to 0.099; and for metaphor the class effect, at 0.001 to 0.006, is more than an order of magnitude below the interaction at 0.023 to 0.078, though it still clears zero in 6 of 6. This suggests that the three categories differ less in whether they are represented than in how much of what distinguishes them is carried by a direction shared across words rather than word by word.

Both the class effect and the interaction grow stronger through the layers of the model. The interaction is essentially absent at layer 0 in all three

²Almost every item is a single subword, so this choice has little bearing on the results: zero items are split into multiple subword tokens under Pythia’s tokenizer, and only eight are split into multiple subword tokens under OPT-BabyLM’s tokenizer.

categories, between 0.0004 to 0.0044 across every category and model, and rises to between 0.0228 to 0.2450 (Figure 4, top row). This is not surprising since at the first layer, the model has yet to integrate enough context into the representation to commit to the class of the representation.

Turning to how those components are oriented, the class direction carries almost none of the item effect. At the deepest layer, item-class overlap runs 0.0032 to 0.0067 for part of speech and 0.0025 to 0.0062 for grammatical role, so projecting out the class direction leaves more than 99% of the item effect intact; metaphor’s class effect is too small to fit a subspace to, so there is no orientation to read for it. While both overlaps are technically less orthogonal than a random subspace, the overlap itself is tiny, so the two are near-orthogonal rather than exactly orthogonal.

In contrast to the main effects, the interaction is not orthogonal to class or item. Across the six models, 0.096 to 0.239 of the interaction for part of speech is in the span of the two main effects, for grammatical role 0.084 to 0.179 is, and for metaphor 0.128 to 0.267 is, against nulls whose upper ends run 0.023 to 0.058, 0.014 to 0.038 and 0.020 to 0.060 respectively. The smallest of these overlaps exceeds the largest of the nulls, so the pattern holds in every model. In other words, the small departure from orthogonality between item and class becomes a large one for the interaction.

4 Experiment 3: Decoding the Interaction

Experiment 2 shows that the interaction is large and that it replicates across independent splits of a word’s occurrences, but neither shows what it is *about*. An item-by-class term captures any reliable difference between a word’s uses in the two classes, and some of that difference is context correlated with the label rather than the label itself, since metaphorical uses occur in different sentences from literal ones and subjects appear in different constructions from objects. So γ is an upper bound on class-binding, and the question is whether any of it is the class.

4.1 Method

We test it by decoding. If γ encodes the class, the class should be recoverable from the part of a token’s representation that is neither the item effect nor the shared class direction, and we remove both explicitly. All decoding happens *within* a single

class	model	accuracy	null
part of speech	BabyLM 1.3B	0.93 [0.91, 0.94]	0.52 [0.49, 0.55]
part of speech	Pythia 1.4B	0.94 [0.93, 0.95]	0.50 [0.48, 0.53]
grammatical role	BabyLM 1.3B	0.85 [0.82, 0.87]	0.50 [0.47, 0.53]
grammatical role	Pythia 1.4B	0.84 [0.82, 0.87]	0.52 [0.49, 0.55]
metaphor	BabyLM 1.3B	0.65 [0.63, 0.66]	0.49 [0.47, 0.50]
metaphor	Pythia 1.4B	0.70 [0.68, 0.71]	0.49 [0.48, 0.51]

Table 2: Decoding the class from the interaction, at the deepest layer of the largest model in each family. The item is held fixed and the shared class direction is projected out, so neither main effect can contribute. *Accuracy* is the mean over them; *null* repeats the decode with the class labels permuted within each form. Brackets are percentile bootstrap intervals over forms, and chance is 0.50.

word, so we ask whether *plant*’s figurative uses can be told from *plant*’s literal ones; the item effect is constant within a word and so cannot contribute. We then estimate β as the difference between the two class means pooled over all items, normalize it, and remove that component from every token, so that whatever separates the classes afterwards is not the direction they share.

What remains is the interaction: the part of a word-at-a-class that is neither a fact about the word nor a fact about the class. We decode it with a cross-validated logistic probe, five folds, classes balanced by subsampling within each word so that chance is 0.50, and average the resulting accuracies unweighted over words, with a word contributing only if it has at least eight tokens in each class. Some words rest on few tokens, so the mean carries a percentile bootstrap interval over words. The whole procedure is then repeated with the class labels permuted within each word, which preserves every word’s token count and class balance while destroying the labeling, and that null is where a probe reading its own cross-validation folds would land. We run this at the deepest layer of the largest model in each family, for all three categories.

4.2 Results

Results are reported in Table 2.

For both families of LLMs across all three linguistic categories, the class is recoverable from the interaction at well above chance while the permuted null sits on chance, reaching 0.94 for part of speech, 0.84 for grammatical role and 0.70 for metaphor, with bootstrap intervals excluding the nulls in all six cases. The null landing on chance is what makes those accuracies interpretable, since it is the same probe, the same folds, and the same

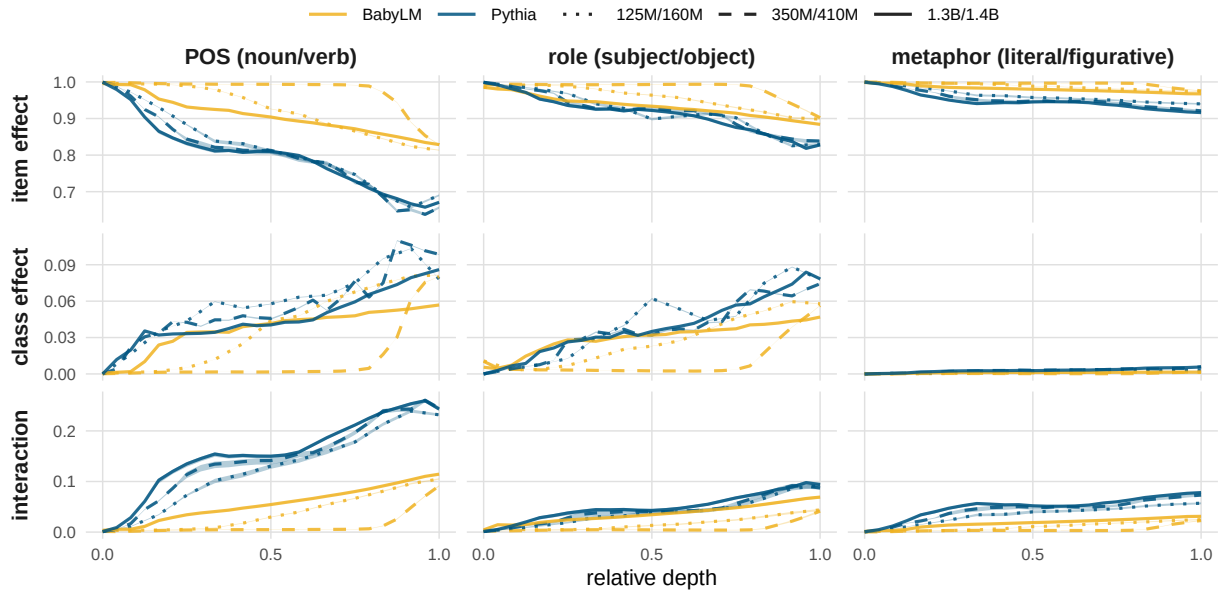


Figure 4: The three components through the network, for every model and linguistic category. Relative depth runs from the static embedding layer (0) to the final block (1). Rows are the three components’ shares of the item-by-class structure, which sum to one within each model and layer. Each line is one model: color distinguishes the training corpus, linetype the three matched widths. The shaded band is that model’s re-split 95% interval.

word-level averaging with only the labeling destroyed.

Further, the effect is graded the same way the interaction’s size is: part of speech decodes best, role next, metaphor least, matching the order in which the interaction’s size falls across the three categories (Table 13). In other words, the geometric and the functional measurement agree on the ranking.

Metaphor is the case where this matters most. Its shared class direction is too small to fit an orientation to (Section 3.4), so a method reading only that direction would take metaphoricity to be barely represented, yet it decodes from the interaction at 0.70 against a null of 0.49. This suggests that most of what distinguishes its uses lives in the interaction rather than in a direction shared across words.

The results demonstrate that γ carries the specific label rather than generic correlated context, since generic context would not survive holding the word fixed and removing the shared axis. It does not show that γ is *only* the label: context correlated with the class within a word survives both removals, so the interaction remains an upper bound.

5 Discussion

Limitations

Everything here is English, and rests on three linguistic categories measured in two model families

at three scales each. The graded ordering we report is an ordering across *these* categories.

The measure is invariant to per-dimension rescaling and to rotation, but not to arbitrary linear reparameterization, so overlap between the item and class effects is relative to the model’s own basis. We take that to be the basis of interest, since LayerNorm and the feed-forward nonlinearities act coordinate-wise and a downstream output layer operates on the unwhitened space; a reader who wants a fully gauge-free quantity will not find one here. The component sizes, their growth with depth and whether their intervals exclude zero do not depend on that choice.

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A Validating the Measure

Experiment 1 plants structure in a *language* and measures a trained model’s representation. A discrepancy there is ambiguous between the measure missing something and the model not mirroring the language, and the second is known to happen: the linear condition cannot represent an interaction at all. This appendix removes the ambiguity by planting μ , α , β and γ **directly in a representation**, with no model and no training, and passing the result to the same function the language models are measured with. Ground truth now sits at the level the measure operates on.

The three components are drawn at random and then centered so each satisfies the constraints its role implies: α and β sum to zero over their own index, γ over both. Without that a planted γ would partly be an item or class main effect and the planted shares would be wrong before the measure saw them. Each is rescaled to hit a target share of the total squared length across the grid, under the same weighting the measure uses. Each mean is then observed n times under isotropic noise, whose scale is set as a multiple of the mean squared length. The grid crosses the planted shares (the same factorial Experiment 1 runs, mapped into share space, plus the three compositions Experiment 2 measures), item counts from 27 to 180, two and four classes, widths from 8 to 2048, 10, 24 and 100 observations per item and class, three noise levels, planted-orthogonal and planted-non-orthogonal effects, and five seeds: 252,000 runs.

At 24 observations per item and class, inside the 10-to-100 range in which the language-model item sets sit, the measured interaction matches the planted one across the whole range (Table 3), and is flagged in essentially every run where anything was planted and about one in twenty where nothing was.

planted	runs	measured	flagged
0.000	6300	0.0000 [0.0000, 0.0094]	5%
0.030	900	0.0323 [0.0199, 0.0418]	100%
0.048	1800	0.0495 [0.0385, 0.0688]	100%
0.056	1800	0.0573 [0.0458, 0.0918]	100%
0.059	1800	0.0607 [0.0495, 0.1041]	100%
0.100	1800	0.1007 [0.0911, 0.1089]	100%
0.111	900	0.1154 [0.1031, 0.1279]	100%
0.167	1800	0.1708 [0.1594, 0.2204]	100%
0.190	1800	0.1937 [0.1825, 0.2736]	100%
0.200	1800	0.2030 [0.1926, 0.2981]	100%
0.240	900	0.2416 [0.2327, 0.2488]	100%
0.333	900	0.3396 [0.3263, 0.3610]	100%

Table 3: Recovery at 24 observations per item and class in 768 or more dimensions, pooled over noise levels, item counts and widths. Brackets are the 2.5th and 97.5th percentiles across runs.

All three effects read essentially zero when nothing is planted, at every sample size, and each is flagged at about the nominal 5% (Table 4): at 24 observations per item and class a true zero class effect reads 0.0000. This calibration is what licenses reading the small metaphor class effect in Section 3.4 as real rather than as the floor: since a true zero reads essentially zero at this sample size, an interval that excludes zero separates even an effect as small as metaphor’s from zero rather than measuring noise.

effect	observations	median	95th pct	flagged
class effect	10	0.00000	0.0081	6.1%
class effect	24	0.00000	0.0034	6.5%
class effect	100	0.00000	0.0009	6.3%
interaction	10	0.00009	0.0644	6.2%
interaction	24	0.00004	0.0309	5.9%
interaction	100	0.00002	0.0083	4.9%
item effect	10	0.00007	0.0568	4.1%
item effect	24	0.00003	0.0255	3.6%
item effect	100	0.00000	0.0064	5.0%

Table 4: What the measure reports when an effect is planted at exactly zero, by observations per item and class, pooled over every other axis of the validation grid.

The rate at which a genuine zero is nonetheless flagged varies with width and effect rather than holding at the nominal 5% (Table 5). The class effect is the most liberal, and most so at the smallest width, where a true zero is flagged 14% of the time; the item effect is the most conservative, falling to 0% at the widest. No setting exceeds ten percent, and pooled across widths the class and interaction are a little more liberal than the item effect. This is the floor the Experiment 1 detections at small width are read against, so a lone flag at $d = 8$ is weak on its own. The language-model results sit at $d \geq 768$, where the rate is mildly liberal at worst, and each rests on the effect clearing zero in all six models, which even an eight-percent per-model rate would not produce by chance.

width d	item	class	interaction
8	3%	14%	7%
16	7%	3%	5%
32	4%	4%	10%
128	6%	4%	5%
768	2%	8%	3%
1024	3%	5%	3%
2048	0%	8%	8%

Table 5: How often a component planted at exactly zero is flagged, by width and by effect.

With the item and class effects planted orthogonal, the measured overlap tracks r/d closely and sits a little below it (Table 6). This is the number Experiment 2’s overlaps are read against: at $d = 2048$ a genuinely orthogonal pair reads 0.00096, and the values reported in Section 3.4 are 2.6 to 7.0 times that.

width d	measured	analytic chance
8	0.17501 [0.06900, 0.40018]	0.25000
16	0.10678 [0.03744, 0.20884]	0.12500
32	0.05609 [0.01864, 0.10454]	0.06250
128	0.01714 [0.00450, 0.02611]	0.02344
768	0.00261 [0.00079, 0.00472]	0.00391
1024	0.00217 [0.00065, 0.00330]	0.00293
2048	0.00096 [0.00031, 0.00165]	0.00146

Table 6: Measured item-class overlap when the two effects are planted orthogonal, at 24 observations per item and class. *Measured* is the median over the runs at that width with the 2.5th and 97.5th percentiles in brackets. The *chance* column is the analytic level of Appendix D.

The orthogonal case shows the overlap does not cry wolf; the planted-non-orthogonal case shows it does not miss. With a known fraction of the class direction placed inside the item subspace, the measured overlap rises well above the chance level and falls outside its own null in the large majority of runs (Table 7): at $d = 2048$ a planted-non-orthogonal pair reads 0.0036 against a chance of 0.00049, detected in 89% of runs. So a reported overlap is one the measure can place at chance when the effects are orthogonal

and lift above it when the non-orthogonality is real, which is what makes the Experiment 2 overlaps interpretable in both directions.

width d	measured	chance	detected
8	0.1688 [0.0824, 0.3998]	0.12500	9%
16	0.0951 [0.0414, 0.2026]	0.06250	22%
32	0.0593 [0.0225, 0.1119]	0.03125	36%
128	0.0213 [0.0062, 0.0517]	0.00781	63%
768	0.0065 [0.0011, 0.0298]	0.00260	83%
1024	0.0056 [0.0008, 0.0331]	0.00098	82%
2048	0.0036 [0.0004, 0.0290]	0.00049	89%

Table 7: Measured item-class overlap when a fraction of the class direction is planted INSIDE the item subspace (non-orthogonal), at 24 observations per item and class. *Measured* is the median over the runs at that width with the 2.5th and 97.5th percentiles in brackets; *chance* is the analytic random-orientation level; *detected* is the fraction of runs whose overlap falls outside that null’s 95% interval.

B Crowding and Orthogonality

Whether the item and class effects come out orthogonal depends on whether the representation had room for them to be, which is a fact about crowding rather than about the language. The languages are held fixed across these runs; only the hidden width changes. Figure 5 shows the overlap against crowding and Table 8 the interaction size and both overlaps at each crowding value.

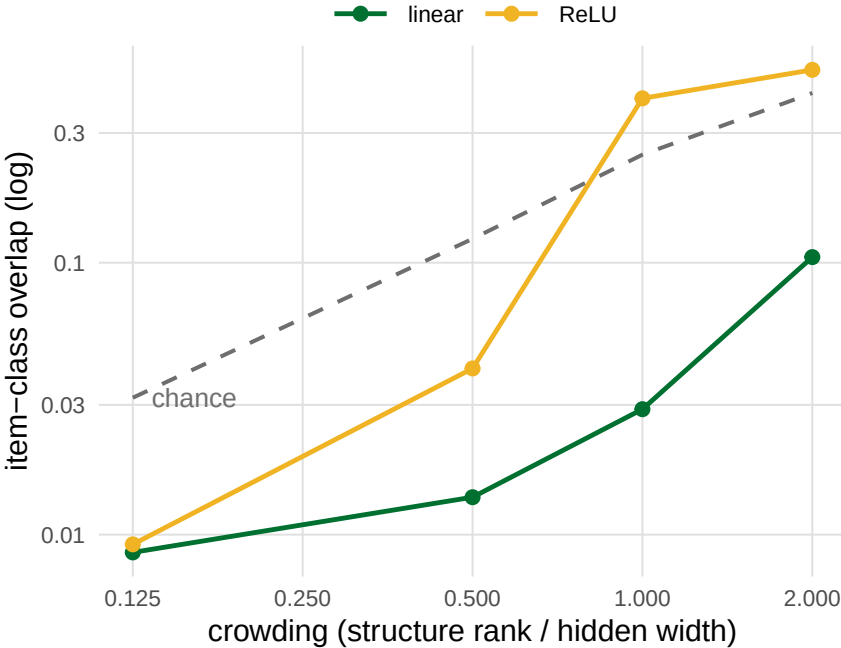


Figure 5: Item-class overlap against crowding, by activation, over the 7,560 artificial-language runs. Crowding is the rank the structure needs divided by the hidden width. The dashed line is the median upper end of the null for an arbitrarily oriented effect. Both axes are logarithmic.

condition	crowding	runs	interaction	item-class overlap	interaction overlap
linear	0.125	405	0.008 [0.002, 0.017] (241/405)	0.0086 (0.4×, 333/405)	—
linear	0.500	405	0.004 [0.001, 0.019] (189/405)	0.0137 (0.2×, 398/405)	—
linear	1.000	405	0.002 [0.000, 0.024] (57/405)	0.0289 (0.2×, 348/393)	—
linear	2.000	396	0.002 [0.000, 0.556] (111/396)	0.1049 (0.4×, 188/319)	—
ReLU	0.125	405	0.083 [0.036, 0.186] (405/405)	0.0092 (0.6×, 225/405)	0.087 (1.3×, 316/405)
ReLU	0.500	405	0.037 [0.003, 0.073] (367/405)	0.0408 (0.7×, 231/405)	0.142 (0.7×, 214/332)
ReLU	1.000	405	0.008 [0.000, 0.046] (237/405)	0.4022 (6.9×, 271/394)	0.230 (1.1×, 40/97)
ReLU	2.000	315	0.003 [0.000, 0.429] (189/315)	0.5119 (5.1×, 122/226)	0.582 (2.4×, 40/77)

Table 8: Interaction size and the two overlaps by activation and crowding, over the 7,560 artificial-language runs. Crowding is the rank the structure needs divided by the hidden width; each crowding value corresponds to one width, from 128 down to 8. *Interaction* is the median share of the representation over the runs in the row, their interquartile range in brackets, and how many have a re-split 95% interval excluding zero. The overlap columns give the median fraction surviving the projection, then in parentheses how many times its own null median that is and how many runs fall outside that null. The linear arm’s interaction stays below the 2% floor at which orientations are reported, so its last column is empty.

Two further properties of the language move the composition, and are given here rather than folded into Figure 3 (Table 9). Context competes with the interaction for room: the more of the language that is context, the smaller the interaction’s share of the item-by-class structure. And the balance the language strikes between the item and the class is reflected in the representation, the measured item share rising with the planted one.

property	level	runs	item	class	interaction
context in the language	under 30%	720	0.451	0.482	0.027
	30 to 70%	572	0.494	0.460	0.016
	over 70%	745	0.505	0.459	0.010
item against class	item < class	694	0.330	0.605	0.020
	item = class	674	0.492	0.447	0.019
	item > class	669	0.643	0.311	0.014

Table 9: How two other properties of the artificial language move the composition of the item-by-class structure, under ReLU, pooled over widths. Rows within a property partition the same runs. Values are medians of the three shares over the runs in the row.

C Model Details

family	model	params	width	layers	training data	tokens
BabyLM	opt-babylm-125m	125M	768	12	BabyLM (child-available)	~2B
	opt-babylm-350m	350M	1024	24	BabyLM (child-available)	~2B
	opt-babylm-1.3B	1.3B	2048	24	BabyLM (child-available)	~2B
Pythia	pythia-160m	160M	768	12	the Pile	~300B
	pythia-410m	410M	1024	24	the Pile	~300B
	pythia-1.4b	1.4B	2048	24	the Pile	~300B

Table 10: The six models, paired by width. Training-token counts are approximate; the OPT-BabyLM models train for 20 epochs over a 100M-word corpus, so their effective token exposure is roughly twenty times the corpus size. *Layers* counts the transformer blocks, and the measure is applied to each block’s output plus the embedding layer.

The six models are listed in Table 10. The two families are matched on width rather than on parameter count, which is what the measure is sensitive to: the chance level for overlap is the class subspace rank over the width (Appendix D), so a comparison across models of different width has to be read against different chance levels. Pairing at 768, 1024 and 2048 removes that from the family comparison.

They differ in two ways that matter elsewhere in the paper. The training corpora differ by more than two orders of magnitude in size, which is the manipulation of interest. And they position tokens differently: Pythia uses rotary embeddings, applied inside attention, while OPT learns absolute position embeddings added at the input. Appendix F uses that difference to rule out absolute position as the source of the interaction.

D The Decomposition

D.1 The item-by-class means

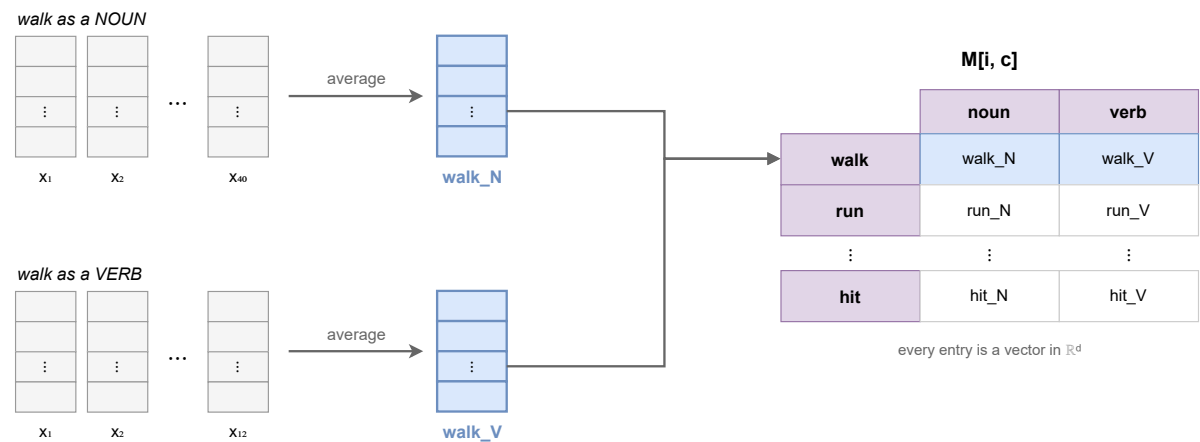


Figure 6: Each word in each class is observed in many contexts, and those observations are averaged within the pairing to give one of the item-by-class means. Every one of them is a vector in $\mathbb{R}^d$, and every item appears in every class.

A word used in a given class is observed in many contexts, and the representation differs a little in each. Averaging those observations within the pairing gives one of the item-by-class means: for items $i \in \{1, \dots, I\}$ crossed with classes $c \in \{1, \dots, C\}$, $M[i, c] \in \mathbb{R}^d$ is the mean representation of item i in class c (Figure 6). An item observed in only one class is dropped, as is an item with fewer than 10 tokens in either class, so the set is complete and every mean rests on a comparable number of observations. Because every item appears at every class, the means arrange into I by C positions, and the rest of this appendix calls that arrangement the grid when what matters is the positions rather than the vectors sitting at them. The variation across contexts that the averaging sets aside is not discarded; it is what the two splits of Section 2.2.2 are drawn from.

D.2 The decomposition as a projection

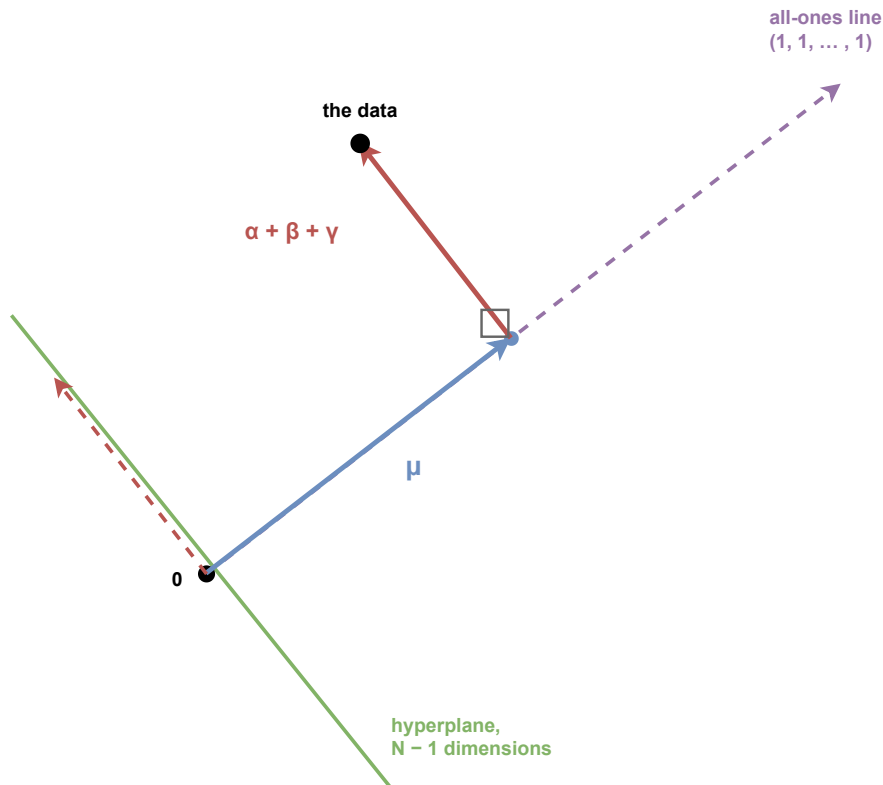


Figure 7: Centering, drawn for a single hidden dimension read across the whole grid. The all-ones direction is where every position takes the same value, so it carries no differences and no variance, and μ is the projection onto it. Removing it leaves a hyperplane of dimension $N - 1$, and every remaining component is a displacement inside it. The dashed copy is that displacement moved to the origin, where it lies in the hyperplane.

Two spaces are in play. The representations themselves live in $\mathbb{R}^d$. The decomposition happens in the grid index space: fix one hidden dimension, read it across all IC positions of the grid, and it is a vector in $\mathbb{R}^{IC}$ (Figure 7). Everything below is that operation carried out once per hidden dimension, with the results stacked back up into the vectors α_i , β_c and γ_{ic} .

In that space the all-ones direction is the one in which every position of the grid takes the same value. Moving along it adds the same amount everywhere and creates no difference between any two positions, so it is the direction along which nothing distinguishes anything, and μ is the projection onto it. Removing it leaves the orthogonal complement, and the three remaining components are displacements inside that complement rather than raw values.

Each component is fixed by two choices, one per axis: along the item axis either average or center, and along the class axis the same. Averaging keeps what is common and discards what makes positions differ along that axis; centering does the reverse. The four combinations are the four components of Equation 2:

	varies over items	varies over classes
μ	no	no
α_i	yes	no
β_c	no	yes
γ_{ic}	yes	yes

So α_i is what holds of an item in both classes, β_c is what holds of a class for every item, and γ_{ic} is what varies along both at once, which is why it cannot be a fact about either one alone. Since averaging

and centering along an axis add back to what was there, the four reconstruct $M[i, c]$ exactly, and they are mutually orthogonal.

Centering an axis costs one dimension, the all-ones direction along it. The four components therefore occupy 1, $I-1$, $C-1$ and $(I-1)(C-1)$ of the grid's IC positions, which is why two classes leave β a single direction whatever I is. Nothing is left over as error: γ is not a residual in the sense of a model that failed to fit, but the part of the representation belonging to the pairing of a particular item with a particular class.

Two consequences are used in the body. First, $\sum_i \alpha_i = 0$, $\sum_c \beta_c = 0$ and $\sum_i \gamma_{ic} = \sum_c \gamma_{ic} = 0$ follow from the centering rather than being imposed as constraints. Second, an **additive** set of means, one of the form $M[i, c] = A(i) + B(c)$, has $\gamma = 0$ *identically*: writing $\bar{A}$ and $\bar{B}$ for the averages, $\mu = \bar{A} + \bar{B}$, $\alpha_i = A(i) - \bar{A}$, $\beta_c = B(c) - \bar{B}$, and the remainder cancels exactly. This is the algebra behind the linear condition of Experiment 1: its population means, the pairing means taken over the same context distribution everywhere, are additive and its interaction is zero. What it is measured on is the means of its observed sample, which can also hold a trace of each pairing's own contexts (Experiment 1, Results), so the floor the linear arm provides is read off the planted-zero languages, not deduced.

D.3 Sizes

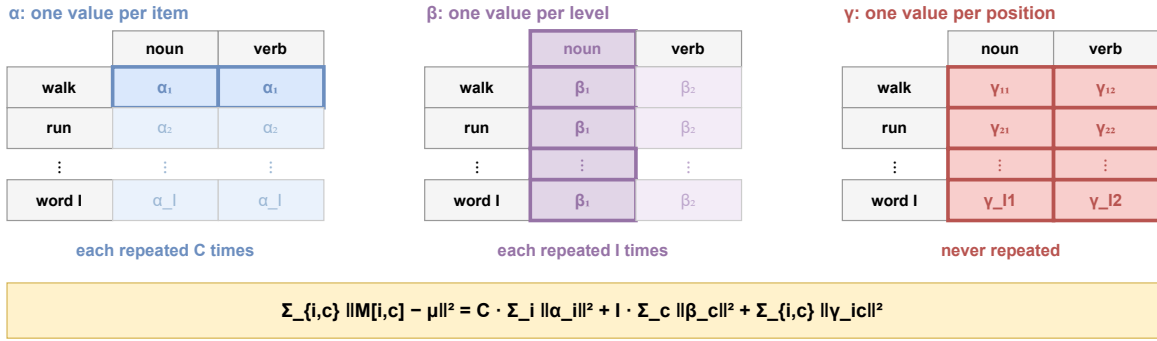


Figure 8: How many positions of the grid each component occupies. α_i does not depend on the class, so it occupies all C positions of its row; β_c does not depend on the item, so it occupies all I positions of its column; γ_{ic} takes its own value at every position and is never repeated.

Because the three components are mutually orthogonal, their squared lengths add rather than overlapping, and the total squared length across the grid partitions exactly:

$$\sum_{i,c} \|M[i, c] - \mu\|^2 = C \sum_i \|\alpha_i\|^2 + I \sum_c \|\beta_c\|^2 + \sum_{i,c} \|\gamma_{ic}\|^2,$$

C and I are not a choice of scaling. They are how many positions of the grid each component occupies (Figure 8): α_i does not depend on the class, so it sits at all C positions of its row, and β_c does not depend on the item, so it sits at all I positions of its column. Every one of the I items is displaced by the same β_c , so the displacement present in the grid is I copies of it. The interaction is simply summed because it takes its own value at every position and is never repeated. Dividing all three by IC would report averages per position instead of totals and leave every share unchanged; what the occupancy counts rule out is weighting the three differently from one another.

Sizes are reported as shares of the total. All three terms scale linearly in I , so the shares do not depend on how many items there are and are comparable across categories of different size.

The partition above is of the item-by-class structure, which has already averaged the contexts away, so it leaves open how much of the representation that structure is. Splitting each observation into its pairing mean and its deviation from it, and using the fact that deviations from a mean sum to zero, gives a second exact partition one level up:

$$\sum_x \|x - \bar{x}\|^2 = \underbrace{\sum_{i,c} \sum_{x \in (i,c)} \|x - \hat{M}[i, c]\|^2}_{\text{within a pairing}} + \underbrace{\sum_{i,c} n_{ic} \|\hat{M}[i, c] - \bar{x}\|^2}_{\text{between pairings}}.$$

The raw between fraction overstates the item-by-class structure, because $\hat{M}[i, c]$ is itself an average of n_{ic} observations and carries $d\sigma^2/n_{ic}$ of context noise that lands in the between term. The bias falls as tokens per item rise, so it differs across categories and would not be comparable between them. We therefore report the variance-component form: the within mean square estimates the context variance, the between mean square estimates that plus n_0 times the true spread of the pairing means with n_0 the effective group size of the unbalanced design, and the difference gives the spread alone. Planting no structure at all, the raw fraction reads 0.081 at 12 observations per pairing and 0.005 at 200, while the corrected one reads zero at both.

Because the corrected quantities are variance components rather than weighted sums, they compose with the sizes: each size times that structure’s share gives that component’s share of the representation, and the three plus the context term account for it entirely.

D.4 The overlaps and their chance level

The sizes say how much of the representation each component accounts for and nothing about direction. Two components can both be large whether or not they occupy the same directions, and separability is the second question: a word’s representation is an item part plus a class part only if the two can be varied independently, which requires them to lie on distinct axes. That question lives in $\mathbb{R}^d$, not in the grid index space the sizes were computed in, and unlike the sizes it does not partition, because each overlap is a relation between two particular components rather than a share of one whole.

Each overlap projects one component into the subspace another occupies:

projected into the subspace of	
α	β
β	α
γ	α and β jointly

The first two are not redundant with each other. One direction and thirty-odd are very different targets to land in by chance, and the chance level below is what puts them on a comparable footing.

Subspaces are orthonormal bases from the leading right singular vectors, with rank set by the participation ratio $r = \text{round}((\sum_k \lambda_k)^2 / \sum_k \lambda_k^2)$ of the scatter eigenvalues, capped at the algebraic maxima above. The participation ratio counts directions discounted by how much each carries: it returns 1 when all the spread lies along one direction and k when it is even across k of them. With two classes β spans one direction, so $r_{\text{class}} = 1$ throughout Experiment 2.

Writing S for the $d \times r$ basis and V for the component being projected, the overlap is $\|VS\|_F^2 / \|V\|_F^2$. V is complete, every vector of the component at full dimension; only S is rank-truncated. So the rank enters through the target alone, which is why it fixes the chance level without altering any measured overlap.

The overlap is a squared cosine. For one vector and one direction, $\langle v, u \rangle^2 / \|v\|^2 = \cos^2 \theta$ exactly, so the ratio is a length-weighted mean of $\cos^2 \theta_i$ over the vectors of V , with θ_i the angle between vector i and the nearest direction in S . Weighting by squared length keeps items with small, poorly determined effects from moving the number. The square root inverts to an angle, which is the more legible scale near the floor: an overlap of 0.19 is 64° and one of 0.0056 is 86° , nearly perpendicular rather than merely small.

Squaring rather than summing the signed dot products is not cosmetic. Because $\sum_i \alpha_i = 0$, the signed version is $\langle \sum_i \alpha_i, \beta \rangle = 0$ identically, whatever the geometry: items leaning one way along the class axis cancel those leaning the other. Squaring is also what makes the two directions of a component count

equally, which is what the question requires, since projecting the class direction out costs an item the same either way.

The chance level is not zero. For a randomly oriented V ,

$$\mathbb{E}[\|VS\|_F^2] = \text{tr}(V^\top V)/d \cdot r \implies \mathbb{E}[\text{overlap}] = r/d,$$

whatever V 's rank or shape. At $d = 2048$ that is 0.0005 for a one-direction class subspace, and Appendix A recovers it empirically, a planted-orthogonal pair landing a little below its own r/d at every width. Any overlap has to be read against that number, which moves by two orders of magnitude across the widths used in this paper.

The measure is exactly invariant to per-dimension rescaling, which the standardization absorbs, and to orthogonal rotation of already-standardized data. It is **not** invariant to general linear reparameterization: we do not whiten, because whitening would make a linear representation read separable in any basis and would discard the basis the model actually computes in. Overlap values are therefore basis-relative, while the sizes and their growth with depth do not depend on the choice.

E Deepest-Layer Measurements

Table 13 gives the deepest-layer values behind the curves of Figure 4, together with the null each overlap is read against.

class	model	item-class overlap	interaction overlap	class effect	interaction
part of speech	BabyLM 125M	0.0056 (4.4×)	0.172 (3.5×)	0.082 [0.081, 0.082]	0.105 [0.104, 0.106]
	BabyLM 350M	0.0052 (5.4×)	0.169 (4.7×)	0.080 [0.080, 0.080]	0.091 [0.090, 0.093]
	BabyLM 1.3B	0.0032 (6.7×)	0.096 (4.5×)	0.057 [0.057, 0.057]	0.114 [0.113, 0.116]
	Pythia 160M	0.0058 (4.6×)	0.230 (4.2×)	0.078 [0.078, 0.078]	0.232 [0.229, 0.234]
	Pythia 410M	0.0067 (7.1×)	0.239 (6.1×)	0.099 [0.098, 0.099]	0.245 [0.242, 0.248]
	Pythia 1.4B	0.0056 (11.6×)	0.209 (9.8×)	0.086 [0.086, 0.086]	0.243 [0.240, 0.246]
grammatical role	BabyLM 125M	0.0038 (3.1×)	0.121 (3.6×)	0.058 [0.058, 0.058]	0.044 [0.042, 0.046]
	BabyLM 350M	0.0038 (4.2×)	0.111 (4.7×)	0.056 [0.056, 0.057]	0.041 [0.039, 0.044]
	BabyLM 1.3B	0.0025 (5.4×)	0.084 (5.7×)	0.047 [0.047, 0.047]	0.069 [0.066, 0.072]
	Pythia 160M	0.0049 (3.9×)	0.179 (5.3×)	0.078 [0.078, 0.078]	0.092 [0.088, 0.096]
	Pythia 410M	0.0047 (5.0×)	0.160 (6.6×)	0.074 [0.074, 0.075]	0.087 [0.084, 0.091]
	Pythia 1.4B	0.0062 (12.5×)	0.141 (11.1×)	0.078 [0.078, 0.079]	0.094 [0.090, 0.098]
metaphor	BabyLM 125M	—	0.223 (3.9×)	0.002 [0.002, 0.002]	0.025 [0.024, 0.026]
	BabyLM 350M	—	0.225 (5.6×)	0.002 [0.001, 0.002]	0.023 [0.022, 0.024]
	BabyLM 1.3B	—	0.128 (4.5×)	0.001 [0.001, 0.001]	0.031 [0.030, 0.032]
	Pythia 160M	—	0.230 (4.2×)	0.003 [0.003, 0.003]	0.057 [0.055, 0.059]
	Pythia 410M	—	0.267 (7.8×)	0.006 [0.006, 0.006]	0.073 [0.070, 0.075]
	Pythia 1.4B	—	0.249 (13.2×)	0.006 [0.006, 0.006]	0.078 [0.075, 0.080]

Table 13: Deepest-layer measurements, one row per model. The two overlaps are the fraction of the item effect lying in the class subspace and the fraction of the interaction lying in the span of the two main effects, each followed in parentheses by how many times its own rotation null’s median it is. That null holds the effect fixed and rotates the target subspace at random 200 times. The class effect and the interaction are shares of the item-by-class structure, with each model’s re-split 95% interval in brackets.

F Positional Embeddings Are Not the Source

One architectural explanation of the interaction has to be ruled out before it can be read as a fact about lexical representation. OPT-BabyLM adds a *learned absolute position embedding* at the input. The two classes of a linguistic category are not positionally interchangeable in a sentence, since subjects precede objects and nouns and verbs occupy different slots, so a representation that varies with position varies with the class. If that is what the decomposition is picking up, the interaction would be an artifact of where words sit rather than of how they are represented.

We re-extracted every OPT-BabyLM model with *decoder.embed_positions* zeroed and measured it identically. The ablation was verified per model rather than assumed: with the embedding zeroed, the same token at two different offsets must give identical layer-0 states, and it does, in every model (drift 0.0 to machine precision). Pythia has no such module, so there is nothing to zero there.

The right quantity is the class effect’s share of class-relevant structure, the class effect divided by the sum of the class effect and the interaction, which we call the *portable fraction*. If positions were manufacturing the interaction, removing them should leave a larger share of what distinguishes the two classes in a direction shared across words. Absolute sizes cannot be compared here: zeroing the embeddings in a model that was trained with them is a heavy intervention, and it shrinks every component at once, the class effect included.

model	positions	class	interaction	portable fraction
opt-babylm-125m	zeroed	0.012	0.034	0.264
opt-babylm-125m	intact	0.082	0.105	0.439
opt-babylm-350m	zeroed	0.008	0.023	0.253
opt-babylm-350m	intact	0.080	0.092	0.468
opt-babylm-1.3B	zeroed	0.009	0.035	0.198
opt-babylm-1.3B	intact	0.057	0.115	0.333

Table 14: Trained OPT-BabyLM at the deepest layer, with absolute position embeddings intact and zeroed. Sizes are shares of the item-by-class structure and the portable fraction is the class effect divided by the sum of the class effect and the interaction.

Removing the position embeddings makes these representations *less* separable, not more. The portable fraction falls in every model, from 0.439 to 0.264, from 0.468 to 0.253, and from 0.333 to 0.198 (Table 14). Whatever the absolute position embedding contributes, it is contributing to the part of class structure that *is* carried by a shared direction. Take it away and what remains is more word-specific, not less. The interaction is not an artifact of position.

A second line of evidence points the same way and needs no ablation at all. Pythia never had an absolute position embedding to remove, and its portable fractions are 0.253, 0.287 and 0.262, at or below every intact OPT-BabyLM value. The family without the mechanism shows the phenomenon at least as strongly as the family with it. An explanation resting on absolute position embeddings would have to account for the models that do not have them.

We tested this for part of speech, the category where the positional worry is sharpest because noun and verb uses of a form occupy systematically different sentence positions. Grammatical role carries the same concern for the same reason, and behaves the same way. Metaphor does not, since literal and figurative uses of a word are not positionally distinguished.